

Supplementary material

S1. Data provenance

This retrospective secondary analysis used the BigP3BCI version 1.0.0 public archive. The downloaded archive had SHA256 digest eea294aa34e9ed11e5a25d07e30aee9df8b2d467a8309e2c38405a289afcd72f. Before any signal was processed, the ingestion pipeline checked this archive digest, read the distributor checksum manifest, selected non-AppleDouble European Data Format files, and checked every selected file against its manifest digest. The source archive and cache are not redistributed in this package.

All 20 documented source studies were examined. Every study supplied the 16 shared EEG channels sampled at 256 Hz together with the StimulusBegin, StimulusType and PhaseInSequence event channels, so no study was excluded for montage or event-channel incompatibility. Study-scoped participant identifiers were retained to prevent accidental cross-study linkage; they do not identify unique people across studies.

S2. Predictor and outcome reconstruction

The predictor used Train-phase files only. A calibration event was a rising StimulusBegin transition during phase 2, labelled by StimulusType. The 16 shared channels were filtered from 0.5 to 30 Hz with a fourth-order zero-phase Butterworth filter, epoched from 200 ms before to 800 ms after the event, baseline corrected to the prestimulus interval, and rejected at an absolute amplitude above 150 microvolts. A session entered the analysis only when at least 10 target and 40 non-target epochs survived rejection.

Epochs were decimated by taking every twelfth sample. At 256 Hz this leaves 22 samples per channel and 352 features, and an effective sampling rate near 21.3 Hz. The effective rate is be-

low twice the 30 Hz filter edge, so frequencies between about 10.7 and 30 Hz are not represented independently in the feature vector. The primary score is a cross-validated discriminability measure rather than a spectral estimate, and the shrinkage linear discriminant comparator computed from the identical features gave an almost identical result, but this remains a deviation from the documented decimation target and is reported rather than corrected.

The number of cross-validation folds was the smaller of five and the number of Train files in the session. Fold count therefore differs between sessions and between studies, so the sampling precision of the predictor is not identical across the cohorts whose transportability is being compared.

For each Test-phase transition to phase 3, the intended character was the final nonzero CurrentTarget in the directly preceding phase-2 interval and the selected character was the modal nonzero SelectedTarget during phase 3. A trial was eligible when feedback was displayed, both values could be recovered, and FakeFeedback did not override the selection. Correctness required an exact character match.

Table S1. Feedback-phase reconstruction by source study. Eligible counts are at the feedback-phase level. Records additionally require a usable calibration feature set for the participant-session, which is why 19,611 selections enter the analysis rather than 19,688.

Source study	Reconstructed	Eligible	Dominant exclusion reason
Study A	1,404	1,404	none
Study B	858	858	none
Study C	5,680	0	artificial feedback override
Study D	1,230	1,230	none
Study E	240	240	none
Study F	1,079	1,067	feedback not displayed

Source study	Reconstructed	Eligible	Dominant exclusion reason
Study G	1,198	1,198	none
Study H	1,926	1,926	none
Study I	948	948	none
Study J	1,812	1,812	none
Study K	480	480	none
Study L	990	990	none
Study M	1,260	1,260	none
Study N	480	480	none
Study O	1,202	1,187	feedback not displayed
Study P	2,263	0	intended character not recoverable
Study Q	3,888	1,944	feedback not displayed
Study R	1,440	1,440	none
Study S1	360	360	none
Study S2	864	864	none
Total	29,602	19,688	

S3. Model specification

The development model was a logistic regression of character-level correctness on the standardised calibration score, fitted on character-expanded records from the development studies, with standardisation using development-study means and standard deviations only. For a session with calibration score s , the estimated accuracy is the inverse logit of $a + b (s - m) / d$, where m and d are the development mean and standard deviation of the score.

S4. Uncertainty

Two uncertainty statements are reported and are not interchangeable.

The participant-cluster bootstrap used 2,000 deterministic replicates. In each replicate, development participant clusters were resampled within source study, the model was refitted, and withheld participant clusters were resampled. The resulting intervals describe uncertainty conditional on the observed set of source studies, because the same studies appear in every replicate by construction.

The study-level summary treats the source study as the unit of replication. The withheld-cohort estimates were summarised by their mean and between-study standard deviation, with a t-distributed interval for the mean on $k - 1$ degrees of freedom and a prediction interval for an unrepresented cohort computed as the mean plus or minus t times the between-study standard deviation times the square root of one plus one over k .

Table S2. Study-level summaries.

Quantity	Mean across cohorts	Between-cohort SD	95% interval for the mean	95% interval for an unrepresented cohort
Mean absolute error	0.104	0.048	0.080 to 0.128	0.001 to 0.208
Brier skill score	0.167	0.210	0.063 to 0.271	-0.287 to 0.621
Character-weighted AUC	0.710	0.105	0.658 to 0.762	0.482 to 0.937
Calibration intercept	-0.078	1.073	-0.612 to 0.456	-2.405 to 2.249
Calibration slope	1.167	0.586	0.875 to 1.458	-0.104 to 2.437

S5. Association at three levels

Table S3. Association between calibration discriminability and observed accuracy.

Level	n	Pearson r	Spearman rho
Sessions, pooled	410	0.699	0.700
Sessions, centred within cohort	410	0.652	0.628
Participants	271	0.701	0.749
Preceding session to later session	139 pairs	0.525	0.440
Same session, reference for the row above	139 pairs	0.702	0.620

Within-cohort Pearson r ranged from 0.083 to 0.920 with a median of 0.630 and was computed in all 18 contributing cohorts.

Session accuracy clustered within participant with an intraclass correlation of 0.412 across 410 sessions in 271 participants, giving approximately 338 effective independent sessions. The predictor is constant within a session, so the 739 session-condition records carry 410 distinct predictor values.

S6. Sensitivity analyses

Table S4. Prespecified sensitivity analyses. Each analysis re-runs the complete withheld-cohort procedure on the retained data.

Analysis	Cohorts	Mean absolute error	Between-cohort SD	Interval for an unrepresented cohort	Calibration-slope SD
Primary, all contributing cohorts	18	0.104	0.048	0.001 to 0.208	0.586
Cohorts with outcome SD at least 0.10	12	0.129	0.042	0.034 to 0.224	0.674
Records with artifact rejection at most 20%	18	0.084	0.033	0.012 to 0.157	0.438

Analysis	Cohorts	Mean absolute error	Between-cohort SD	Interval for an unrepresented cohort	Calibration-slope SD
Records with at least 10 eligible selections	18	0.104	0.048	0.001 to 0.207	0.586
Cohorts without a documented ALS population	14	0.104	0.052	-0.012 to 0.219	0.716
ALS cohorts, prespecified primary subgroup	4	0.101	0.025	0.013 to 0.189	0.204

Artifact rejection had a median of 0.002 across records, but 110 of 739 records exceeded 20% and the maximum was 0.97. The analysis restricted to records at or below 20% rejection is the only one in which both estimation error and calibration-slope variability improved.

Observed session-condition accuracy was at 100% in 228 of 739 records (30.9%). Three cohorts had mean accuracy at or above 0.96, where there is little variation to estimate.

S7. Comparator predictors

Comparator scores were computed from the identical calibration epochs and evaluated under the identical withheld-cohort protocol: regularised linear discriminant analysis area under the curve, grouped cross-validated classification accuracy, mean target-minus-non-target amplitude at Pz between 250 and 500 ms, the same contrast averaged over six posterior channels, and the maximum posterior signed r-squared.

S8. Reproducibility

The analysis pipeline executes archive validation, source metadata extraction, feedback-phase reconstruction, calibration feature extraction, withheld-cohort validation, the widened-design analyses, and sensitivity analyses. Frozen outputs are written to output/expanded/: study_inventory.csv, analysis_records.csv, external_validation_metrics.csv, external_validation_predictions.csv, random_effects_pooling.csv, als_subgroup_metrics.csv, transfer_to_als.csv, als_moderation.csv, null_benchmark.csv, within_study_association.csv, participant_level_association.csv, across_session_association.csv, across_session_pairs.csv, session_clustering.json and sensitivity_analyses.csv. The repository test suite contains 41 tests covering provenance, European Data Format parsing, event reconstruction, feature extraction, validation, the widened-design analyses, and rendering.

S9. Transparency statement

The study involved no new data collection, participant contact, prospective enrolment, or intervention. It does not establish a diagnostic, prognostic, causal, or treatment effect. Character-level online selection accuracy is an operational endpoint and should not be presented as communication success, quality of life, or a clinical outcome. The source studies vary in protocol, and the archive does not permit cross-study person-level linkage. Four source studies carry a documented ALS population; the remaining cohorts are described as other cohorts because the documentation does not support a positive characterisation, and no participant-level clinical characteristics were available.